
W H I T E P A P E R

Multi-Horizon Stock Return Prediction Using Heterogeneous Data Modalities:

A Comparative Study of LightGBM, Temporal Fusion Transformer, and
iTransformer on GOOGL

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github.com/hamkhadafi/googl-stock-prediction

Abstract

This paper presents an end-to-end machine learning pipeline for multi-horizon forward return prediction of Alphabet Inc. (GOOGL) stock, comparing three architecturally heterogeneous models: LightGBM (gradient-boosted decision trees), Temporal Fusion Transformer (TFT), and iTransformer. The pipeline integrates three data modalities—technical indicators, fundamental financial ratios derived from SEC EDGAR XBRL filings, and macroeconomic indicators from the FRED API—into a 157-column feature matrix spanning 3,134 daily observations (2014–2026). Feature engineering includes 70 technical indicators, 47 fundamental ratios, 26 macro features, and 6 valuation metrics, with availability masks for sparse fundamental columns. Model evaluation employs a walk-forward temporal split (train: 2014–2021, val: 2022–2023, test: 2024–June 2026) with non-overlapping rebalance windows to prevent lookahead bias. On the test set, iTransformer achieves the highest directional accuracy at 1-day (51.4%) and 5-day (55.9%) horizons and the highest Sharpe ratios at these horizons (1d: 1.09, 5d: 1.83), while LightGBM produces the lowest MAE across 1-day and 5-day horizons and dominates at 20-day. No model consistently beats a buy-and-hold benchmark on absolute return during the 2024–2026 test period—a strong GOOGL bull run—which is presented as an expected and informative outcome rather than a failure. This paper is an academic research exercise and does not constitute financial advice.

Keywords—stock return prediction, multi-horizon forecasting, LightGBM, Temporal Fusion Transformer, iTransformer, feature engineering, walk-forward validation

1. INTRODUCTION

Predicting asset returns is one of the most studied problems in quantitative finance and machine learning. The challenge is fundamentally difficult: financial markets aggregate information rapidly, and any predictable signal tends to be arbitrated away. Yet the engineering challenge—building a pipeline that correctly handles heterogeneous data sources, temporal dependencies, and the many forms of lookahead bias that make backtests misleading—remains substantial and practically valuable independent of the strength of the prediction signal.

This project approaches stock return prediction primarily as an ML engineering problem. The central question is not “can we reliably predict GOOGL returns”—prior literature suggests the answer is at best “weakly, at short horizons”—but rather: how do fundamentally different model architectures behave on the same feature set and evaluation protocol, and what do their differences reveal about the nature of the prediction task?

Three models are compared across three prediction horizons (1-day, 5-day, 20-day):

- LightGBM: a gradient-boosted decision tree ensemble that handles tabular features natively, is computationally efficient, and serves as a strong tabular baseline.
- Temporal Fusion Transformer (TFT): an attention-based multi-horizon forecaster designed for time series with mixed static and dynamic covariates, offering interpretable variable selection.
- iTransformer: an inverted Transformer architecture (Liu et al., ICLR 2024 [1]) that applies attention across variates rather than timesteps, implemented from scratch in PyTorch.

The contribution of this work is a reproducible, end-to-end pipeline with principled design decisions at each stage: multi-source data integration, stationarity-aware feature selection, walk-forward

validation with non-overlapping rebalance windows, and honest reporting of results including underperformance relative to buy-and-hold.

2. DATA

2.1 Sources and Coverage

Three data modalities are collected from publicly accessible sources:

Table 1. Data Sources

Source	Data	Coverage	API Key
Yahoo Finance (yfinance)	Daily OHLCV prices	2010–present	No
SEC EDGAR XBRL API	Quarterly financial statements	2012–present	No
FRED API	Macroeconomic indicators	2010–present	Yes (free)

The training window begins in 2014 rather than 2010 to allow a fundamental rolling lookback warmup period following Alphabet’s first available EDGAR XBRL filing. This yields a feature matrix of 3,134 daily rows. News and sentiment data were evaluated (Finnhub free tier, GDELT Project) but excluded: Finnhub covers only approximately two years of history on the free tier, and GDELT’s Doc 2.0 API maintains a three-month rolling window, neither sufficient for a 2014–2026 training window.

2.2 Temporal Split

A single walk-forward split is used: training (2014–2021), validation (2022–2023), and test (2024–June 2026). This split respects the strict temporal ordering required for financial time series. No random shuffling or k-fold cross-validation is applied, as these would introduce lookahead bias.

3. FEATURE ENGINEERING

3.1 Feature Groups

The final feature matrix contains 157 columns: 5 raw OHLCV inputs, 149 engineered features, and 3 target variables. Features are organized into six groups:

Table 2. Feature Groups

Group	Count	Key Features
Technical	70	Log returns (7 horizons), SMA/EMA ratios, MACD, RSI, Stochastic %K/%D, Williams %R, CCI, ADX, Bollinger Bands, ATR, realized volatility (3 windows), OBV, VWAP, MFI, Donchian and Keltner Channels, Ichimoku
Fundamental	47	Profitability margins, ROE/ROA/ROIC, liquidity and leverage ratios, Piotroski F-Score (9 components), Altman Z-Score (4 components), QoQ growth metrics, R&D intensity
Fundamental sparse	3+3	Raw values for dda, short_term_investments, long_term_debt (>50% missing) + binary availability masks
Macro	26	12 FRED series, yield curve spread (10y–2y), real rate, VIX regime (0/1/2), rate-of-change for daily series, CPI MoM, GDP QoQ, GOOGL vs NASDAQ 20d relative return
Valuation	6	P/E, P/B, P/S, EV/EBITDA, FCF yield, log market cap

3.2 Key Engineering Decisions

Forward-fill strategy. Fundamental (quarterly) data is forward-filled without a day limit, as quarterly filings are valid until the next filing date. Monthly macro series are similarly forward-filled. Daily macro series use `ffill(limit=5)` to avoid propagating gaps beyond one trading week. S&P 500 data is sourced from Yahoo Finance (`^GSPC`) rather than FRED, which showed 25% data gaps on the free tier.

Availability masks. Three fundamental columns (`dda`, `short_term_investments`, `long_term_debt`) have greater than 50% missing values in raw EDGAR quarterly filings. Rather than dropping these columns, binary mask features are added alongside the forward-filled values, allowing models to learn the distinction between a day with a fresh filing value and a day with a propagated stale value. This is a form of informative missing-data encoding.

Two feature lists. After EDA-based redundancy removal (dropping one feature from pairs with $|\text{Spearman } r| > 0.95$) and Augmented Dickey-Fuller stationarity testing, two feature lists are produced. The tree model list (111 features) retains non-stationary level features such as raw SMA, EMA, and valuation ratios, which LightGBM can handle via split thresholds. The sequence model list (89 features) excludes these non-stationary features, as neural architectures are sensitive to distribution shift across train and test periods.

Target construction. Three forward return targets are defined: $\text{target_return_1d} = (\text{Close}[t+1] - \text{Close}[t]) / \text{Close}[t]$, and analogously for 5-day and 20-day horizons. All price data is adjusted for Alphabet's 20-for-1 stock split of July 2022. Outlier clipping at the 1st/99th percentile of the training distribution is applied to all continuous features before saving.

4. MODELS

4.1 LightGBM

Three independent LightGBM models are trained, one per horizon, using the 111-feature tree model list as input and MAE as the validation objective. The architecture is intentionally conservative: `max_depth=4`, `num_leaves=15`, `learning_rate=0.02`, with L1 and L2 regularization (`alpha=0.1`, `lambda=0.5`) and subsampling (`subsample=0.8`, `colsample_bytree=0.7`). Early stopping at `patience=100` on validation MAE, with up to 2,000 boosting rounds.

The best iteration counts—all below 25 rounds (1d: 24, 5d: 19, 20d: 20)—indicate that early stopping triggered very early, a consequence of the weak and noisy return signal rather than misconfigured hyperparameters. With a learning rate of 0.02, 24 rounds is approximately equivalent to a learning rate of 0.48 at one round, which is a very blunt fit. This is consistent with the near-zero R^2 values across all horizons: the model finds a local minimum quickly and cannot improve further without overfitting to noise.

Feature importance by gain is computed per horizon. Top features at 1-day are `obv_momentum`, `vix`, and `vix_20d_chg`—short-term momentum and volatility signals. At longer horizons (5-day and 20-day), fundamental valuation ratios (`pb_ratio`, `ps_ratio`) and macroeconomic variables (`treasury_2y`, `adx_14`) dominate. This shift in feature importance across horizons is itself informative: short-term price movements are driven by market microstructure and sentiment, while longer-term returns show greater sensitivity to valuation and rates.

4.2 Temporal Fusion Transformer

The TFT [2] is an attention-based multi-horizon forecaster that combines per-timestep gate mechanisms (Gated Residual Networks), a Variable Selection Network (VSN) for soft feature selection, and multi-head self-attention for temporal pattern capture. The implementation uses `pytorch-forecasting` with Lightning v2.x (`lightning.pytorch` import path, not the deprecated `pytorch_lightning`). Architecture: `hidden_size=64`, `attention_head_size=4`, `dropout=0.2`, `hidden_continuous_size=32`, encoder length 60 trading days, output 7 quantiles with the median (`q0.5`) used as the point forecast.

Trainable parameters: approximately 1,044,195. At roughly 2,000 training samples, this represents a fundamental capacity-data mismatch. The TFT architecture is designed for datasets typically 5–10× larger; at this scale, the VSN's soft feature selection partially mitigates overfitting, but not sufficiently. The 1-day and 5-day models hit the maximum epoch ceiling (60 epochs) without triggering early stopping, with validation loss still slowly declining at the final epoch—suggesting the model is underfitted and could benefit from extended training. The 20-day model early stops at epoch 23 with validation loss stagnating.

4.3 iTransformer

The iTransformer [1] inverts the standard Transformer tokenization: instead of treating each timestep as a token (with all variates as features), each variate's full time series is treated as a token. Attention is therefore computed across variates rather than across time, learning cross-variate dependencies. This is particularly well-suited to the present setting: 89 variates × 60 timesteps, where inter-feature relationships may carry more signal than temporal autocorrelation in individual features.

The implementation is written from scratch in PyTorch, following Liu et al. [1]. Architecture: `d_model=64`, `n_heads=4`, `d_ff=256`, `n_layers=2`, `dropout=0.1`. Variate embedding: a linear projection

from the 60-timestep series to $d_model=64$ per variate. Aggregation: mean pooling across variate tokens feeds a linear prediction head. Total trainable parameters: approximately 106,113—roughly 10× fewer than TFT.

Training uses Huber loss ($\delta=0.01$), AdamW optimizer, and CosineAnnealingLR scheduler with early stopping patience=10. The 1-day model early stops at epoch 32 with stable validation loss (val/train loss ratio 1.67×, acceptable). The 5-day and 20-day models both early stop at epoch 11 with validation loss diverging from epoch 2–3 (val/train ratios of 1.91× and 2.85×, respectively). This overfitting at longer horizons is attributable to the noisier multi-day targets combined with the limited training set size, and is consistent with TFT’s behavior at the same horizons.

No explicit feature importance is available for iTransformer: the attention mechanism produces a ($batch \times heads \times n_variates \times n_variates$) matrix from which per-variate importance requires aggregation across heads and batches, which was not implemented in this study.

5. EVALUATION PROTOCOL AND RESULTS

5.1 Evaluation Protocol

All models are evaluated on three metric categories:

- Regression metrics: MAE, RMSE, and R^2 on val and test sets. A zero-intelligence naive baseline (predicting the training-set mean return) is included for LightGBM to contextualize regression performance.
- Directional accuracy: fraction of correct sign predictions, benchmarked against the historical base rate (fraction of positive-return days in that split).
- Simulated trading strategy: a long/flat strategy that goes long when predicted return > 0 , otherwise holds cash. Positions are updated every N days using non-overlapping rebalance windows ($N = \text{horizon}$), eliminating the compounding error that would arise from treating each day’s N -day forward return as an independent daily return. Transaction costs of 5 basis points per round-trip trade are applied. Annualization uses $\sqrt{252 / \text{horizon_days}}$ periods per year.

The non-overlapping rebalance design is a critical distinction from naive backtests. For a 5-day horizon model, the strategy holds a position from day t to $t+5$, then rebalances. It does not re-evaluate daily using the 5-day prediction, which would double-count overlapping return windows and artificially inflate Sharpe ratios.

5.2 Regression Metrics

Table 3. MAE — Validation Set

Model	1d	5d	20d
LightGBM	0.01634	0.03670	0.06375
Naive baseline	0.01635	0.03680	0.06422
TFT	0.01784	0.04179	0.07645

Model	1d	5d	20d
iTransformer	0.01635	0.04122	0.07092

Table 4. MAE — Test Set (2024–2026)

Model	1d	5d	20d
LightGBM	0.01384	0.03368	0.07445
TFT	0.01421	0.03738	0.08446
iTransformer	0.01419	0.04380	0.07365

All models produce near-zero or negative R^2 on both val and test sets, indicating that none meaningfully explains variance in forward returns beyond the mean. This is consistent with the efficient market hypothesis at these horizons and data scales. The naive baseline's MAE is nearly identical to LightGBM across all horizons on the val set, underscoring how thin the predictive edge is on return magnitude.

5.3 Directional Accuracy

Table 5. Directional Accuracy (Test Set) vs. Base Rate

Model	1d Acc.	1d Base	5d Acc.	5d Base	20d Acc.
LightGBM	47.0%	55.3%	49.4%	58.9%	64.5%
TFT	46.4%	55.3%	48.5%	58.9%	47.7%
iTransformer	51.4%	54.7%	55.9%	58.2%	62.6%

Directional accuracy at 1-day and 5-day horizons falls below the base rate for LightGBM and TFT—both models lean toward predicting positive returns more often than the market delivers negative ones. iTransformer is the only model to exceed its base rate at 1-day (51.4% vs 54.7% base) while remaining competitive. At 20-day, LightGBM achieves accuracy virtually equal to the base rate (64.5% vs 64.9%), reflecting that its longer-horizon predictions converge to the unconditional positive-return probability without adding directional information.

5.4 Strategy Performance (Sharpe Ratio)

Table 6. Sharpe Ratio — Strategy vs. Buy-and-Hold (Test Set)

Model	1d Sharpe	5d Sharpe	20d Sharpe	Note
LightGBM	0.29	0.84	1.61	Only model beating B&H at 20d

Model	1d Sharpe	5d Sharpe	20d Sharpe	Note
TFT	0.39	0.72	0.98	Lowest across all horizons
iTransformer	1.09	1.83	1.09	Beats B&H at 5d; near B&H at 1d
Buy-and-hold	1.48	1.53	1.53	Strong 2024–2026 bull run

iTransformer achieves a 5-day Sharpe of 1.83, exceeding buy-and-hold (1.53) on the test set—the only model-horizon combination to do so. This result should be interpreted with caution: the 5-day non-overlapping strategy generates only approximately 110 rebalance periods on the test set, and Sharpe estimates at this sample size carry meaningful uncertainty. The result is promising but not statistically conclusive without confidence interval estimation.

No model beats buy-and-hold consistently. The test period (2024–June 2026) is a strong bull run for GOOGL; strategies that go to cash to avoid predicted down-moves also miss the strongest up-moves. This is an expected structural property of a long/flat strategy without short-selling.

6. DISCUSSION

6.1 Architecture Trade-offs

The three models illuminate distinct trade-offs that generalize beyond this specific prediction task:

LightGBM vs neural architectures on small tabular datasets. With approximately 2,000 training samples, LightGBM outperforms both neural models on MAE at 1-day and 5-day horizons. This is consistent with the established empirical finding that gradient-boosted trees dominate neural networks on small-to-medium tabular datasets [3]. The advantage erodes at 20-day, where LightGBM’s top features (`pb_ratio`, `treasury_2y`) are slow-moving signals that both tree and neural models can learn similarly.

TFT capacity-data mismatch. TFT’s approximately 1.04M parameters—roughly 10× iTransformer—produce the weakest results across all metrics. This is a concrete illustration that architectural sophistication does not compensate for insufficient training data. TFT is designed for datasets an order of magnitude larger; at this scale, the Variable Selection Networks add regularization overhead without sufficient signal to learn from.

iTransformer’s directional edge. iTransformer leads on directional accuracy and Sharpe at 1-day and 5-day horizons despite having the worst MAE at 5-day. This reflects a meaningful distinction: MAE measures return magnitude accuracy, while directional accuracy and strategy Sharpe measure sign prediction. iTransformer’s cross-variate attention—which treats each feature’s time series as a token and learns joint patterns across 89 features simultaneously—appears better at detecting the multivariate configurations that precede directional reversals, even when its magnitude estimates are imprecise.

Longer horizons are harder, but for different reasons per model. At 20-day, iTransformer's training shows clear overfitting (val/train loss ratio 2.85× at early stopping), while LightGBM's 20-day performance improves relative to shorter horizons, driven by valuation and macro features with longer predictive lead times. The 20-day signal is learnable by tree-based models using level features; it is less accessible to neural architectures that require stationarized inputs, which remove much of the valuation signal.

6.2 Engineering Highlights

Several pipeline decisions are worth noting as reusable patterns for similar prediction tasks:

- Availability masks for sparse fundamentals: encoding the distinction between a day with a fresh filing value and a day with a forward-filled stale value allows models to explicitly learn that sparse values should be discounted in non-filing periods.
- Two feature lists (tree vs sequence): separating features by stationarity, rather than applying the same preprocessing uniformly, avoids the known failure mode of feeding non-stationary level features to neural architectures while preserving their value for tree models.
- Non-overlapping rebalance windows: a design choice that is rarely implemented correctly in amateur backtests and that meaningfully affects Sharpe ratio estimates for strategies at horizons greater than one day.
- iTransformer from scratch: implementing the architecture from the original paper rather than using a library enforces a deep understanding of the attention inversion mechanism and produces a model with 10× fewer parameters than the library-based TFT while remaining competitive on direction.

6.3 Limitations

- Single ticker. Results on GOOGL are not generalizable without testing on a broader equity universe.
- No hyperparameter search. All results reflect reasonable defaults; systematic tuning could materially improve all models, particularly TFT.
- iTransformer overfitting at 5-day and 20-day horizons. Higher dropout, stronger weight decay, or data augmentation were not explored.
- Single walk-forward split. Multiple rolling windows would provide more robust performance estimates and uncertainty quantification.
- Test period bias. The 2024–2026 test period is a strong bull market for GOOGL. Strategy performance in bear or sideways conditions is unknown.
- Sharpe estimates at 20-day use only approximately 30 non-overlapping blocks on the test set; confidence intervals are not reported and these estimates are noisy.
- News and sentiment excluded due to the absence of a free historical source with sufficient coverage.

7. CONCLUSION

This paper presents a complete ML engineering pipeline for multi-horizon stock return prediction, comparing LightGBM, TFT, and iTransformer on a 157-feature dataset integrating technical, fundamental, and macroeconomic signals. The results confirm that predicting return magnitude is extremely difficult—all models show near-zero R^2 —while directional prediction at short horizons shows marginal but measurable signal, particularly from iTransformer’s cross-variate attention mechanism. LightGBM dominates at 20-day via valuation and macro features; TFT underperforms due to an architectural capacity-data mismatch; iTransformer achieves the highest Sharpe at 1-day and 5-day horizons.

The primary value of this work is not the prediction results themselves, but the pipeline design: multi-source data integration, stationarity-aware feature selection, availability masks for sparse fundamentals, non-overlapping rebalance evaluation, and honest reporting of underperformance relative to buy-and-hold during a strong bull period. These are the engineering properties that transfer across prediction domains.

Future directions include: (1) extending to a multi-ticker universe to assess generalizability; (2) systematic hyperparameter optimization, particularly for TFT; (3) ensemble methods combining LightGBM predictions (strongest on magnitude) with iTransformer predictions (strongest on direction); and (4) incorporating news sentiment once a suitable long-history source becomes accessible.

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